

Notes: Angeletos and Pavan (Econometrica, 2007)

Efficient Use of Information and Social Value of Information

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1 Big picture

- **Question.** How should dispersed private and public information be used in economies with strategic complementarities/substitutabilities and externalities? When is more information socially valuable, and when can it reduce welfare?
- **Core answer.** The social value of information cannot be inferred from whether actions are strategic complements or substitutes alone. It depends on whether the equilibrium uses information efficiently: whether private incentives to coordinate equal the socially optimal degree of coordination.
- **Positive result.** Strategic complementarity tilts equilibrium actions toward public information, creating more aggregate volatility. Strategic substitutability tilts actions toward private information, creating more cross-sectional dispersion.
- **Normative result.** Higher volatility or dispersion is not automatically inefficient. The planner faces a tradeoff between volatility and dispersion, and the efficient use of information depends on the socially optimal degree of coordination α^* .
- **Main classification.** The paper separates economies into three classes: efficient under complete and incomplete information; inefficient only under incomplete information; and inefficient even under complete information.
- **Social value of information.** If the equilibrium is efficient, more private or public information always raises welfare. If the equilibrium degree of coordination is distorted, one type of information can reduce welfare by moving the economy toward a worse mix of volatility and dispersion. If the complete-information equilibrium is itself inefficient, even accuracy can reduce welfare by pushing the economy closer to an inefficient complete-information outcome.
- **Applications.** The paper uses the framework to clarify apparently conflicting conclusions in competitive production, investment complementarities, beauty contests, business-cycle transparency, Cournot competition, and Bertrand competition.
- **Takeaway.** Public information is not intrinsically bad. Private information is not intrinsically good. Their welfare effects depend on the gap between equilibrium and efficient information use.

2 Incremental contribution

- **Moves beyond positive global-games/beauty-contest results.** Prior work established that public information can receive excessive weight in strategic-complementarity games. Angeletos and Pavan ask the normative question: excessive relative to what social benchmark?
- **Introduces an efficiency benchmark with decentralized information.** The planner cannot centralize private signals or communicate information across agents. The planner can only choose how each agent maps his own information into action. This isolates inefficiency in the *use* of information rather than assuming away information decentralization.
- **Defines the socially optimal degree of coordination.** The paper shows that the efficient allocation is characterized by a parameter α^* analogous to the equilibrium degree of coordination α . Comparing α and α^* summarizes whether agents coordinate too much or too little.
- **Separates accuracy from commonality.** Instead of viewing public and private signals as primitives, the paper decomposes information into forecast accuracy σ^{-2} and commonality δ . This is central because public information changes both accuracy and correlation of beliefs.
- **Provides a taxonomy of welfare effects.** The paper gives a map of “what goes when”: welfare effects of information depend on whether $\kappa = \kappa^*$ and whether $\alpha = \alpha^*$, not simply on complementarity.
- **Reconciles conflicting results in the transparency literature.** Morris-Shin beauty contests, investment-spillover models, and new-Keynesian transparency models can all look similar from a positive perspective but differ sharply in welfare because their $\alpha - \alpha^*$ and $\kappa - \kappa^*$ gaps differ.

- **Extends the analysis to multiple applications.** The paper shows how the same framework delivers different welfare conclusions for competitive production, investment spillovers, business-cycle shocks, Cournot competition, and Bertrand competition.

3 Model objects and measurement

3.1 Agents, actions, and payoff interdependencies

There is a continuum of agents. Agent i chooses an action $k_i \in \mathbb{R}$. Let K denote the cross-sectional mean of actions and let σ_k denote the cross-sectional dispersion of actions. Payoffs are quadratic:

$$u_i = U(k_i, K, \sigma_k, \theta),$$

where θ is the payoff-relevant fundamental.

Interpretation: The model is designed to be abstract enough to cover many coordination economies. K captures aggregate alignment. σ_k captures dispersion across agents. The payoff function permits strategic effects through U_{kK} and external welfare effects through U_K , U_{KK} , and $U_{\sigma\sigma}$.

3.2 Equilibrium degree of coordination

The key private strategic object is

$$\alpha \equiv \frac{U_{kK}}{-U_{kk}}.$$

- $\alpha > 0$ means strategic complementarity: agents privately want to align their actions with the aggregate.
- $\alpha < 0$ means strategic substitutability: agents privately want to differentiate from aggregate behavior.
- The paper assumes $\alpha < 1$ to obtain a unique bounded equilibrium in the linear-quadratic setting.

Interpretation: α is the slope of the best response with respect to the mean action. It is the private value of coordination.

3.3 Information structure

The fundamental is normal:

$$\theta \sim N(\mu, \sigma_\theta^2).$$

Each agent observes a private signal and a common public signal:

$$x_i = \theta + \xi_i, \quad y = \theta + \varepsilon,$$

where ξ_i is idiosyncratic noise and ε is common noise.

The public posterior mean is

$$z \equiv E[\theta \mid y] = \lambda y + (1 - \lambda)\mu, \quad \lambda = \frac{\sigma_y^{-2}}{\sigma_y^{-2} + \sigma_\theta^{-2}}.$$

The private posterior mean after observing (x_i, y) is

$$E[\theta \mid x_i, y] = (1 - \delta)x_i + \delta z, \quad \delta = \frac{\sigma_y^{-2} + \sigma_\theta^{-2}}{\sigma_x^{-2} + \sigma_y^{-2} + \sigma_\theta^{-2}}.$$

Interpretation: z is the public-information component. δ measures how much an agent's posterior leans toward common information rather than idiosyncratic private information.

3.4 Accuracy and commonality

Let

$$v_i = \theta - E[\theta \mid x_i, y]$$

be agent i 's forecast error. The paper identifies:

$$\text{accuracy} = \sigma^{-2}, \quad \text{commonality} = \delta,$$

where $\sigma^2 = \text{Var}(v_i)$ and $\delta = \text{Corr}(v_i, v_j)$ for $i \neq j$.

Interpretation: Public and private information are economically different because they have different effects on the commonality of beliefs. Public information raises commonality; private information lowers commonality. Both can improve accuracy.

4 Models and theoretical specifications

4.1 Complete-information equilibrium

If θ is common knowledge, the unique equilibrium has $k_i = \kappa(\theta)$ for all i , where $\kappa(\theta)$ solves

$$U_k(\kappa, \kappa, 0, \theta) = 0.$$

Because U is quadratic,

$$\kappa(\theta) = \kappa_0 + \kappa_1 \theta.$$

Interpretation: $\kappa(\theta)$ is the complete-information equilibrium target. Under incomplete information, agents choose actions that forecast this target while also taking into account how others forecast it.

4.2 Incomplete-information equilibrium

Proposition 1 shows that equilibrium satisfies

$$k(x, y) = E[(1 - \alpha)\kappa(\theta) + \alpha K(\theta, y) \mid x, y],$$

with the unique linear form

$$k(x, y) = \kappa_0 + \kappa_1 [(1 - \gamma)x + \gamma z],$$

where

$$\gamma = \delta + \frac{\alpha\delta(1 - \delta)}{1 - \alpha(1 - \delta)}.$$

Interpretation: γ is the equilibrium weight on public information. If $\alpha = 0$, $\gamma = \delta$ and agents use information as Bayesian forecasters. If $\alpha > 0$, $\gamma > \delta$: complementarity makes public signals especially attractive because public signals forecast others' actions. If $\alpha < 0$, $\gamma < \delta$: substitutability makes private information more useful because agents want to differentiate.

4.3 Volatility and dispersion

Incomplete information creates two forms of nonfundamental variation:

$\text{Var}(K - \kappa)$ aggregate volatility from common noise,

$\text{Var}(k - K)$ cross-sectional dispersion from private noise.

Interpretation: Public noise moves many agents together and generates aggregate volatility. Private noise moves agents apart and generates dispersion. Complementarity shifts the economy toward the first type of noise; substitutability shifts it toward the second.

4.4 Efficiency benchmark

The efficient allocation is the strategy $k : R^2 \rightarrow R$ that maximizes ex ante welfare subject to decentralized information:

$$E U(k(x, y), K(\theta, y), \sigma_k(\theta, y), \theta).$$

The planner cannot pool private information. The planner can only choose how each agent's own information maps into action.

Interpretation: This benchmark is the central methodological innovation. It avoids comparing equilibrium with a full-information planner. Instead, it asks whether agents use their actual decentralized information in the socially right way.

4.5 First best and efficient allocation

Let $W(K, \sigma_k, \theta)$ denote utilitarian welfare. The first-best complete-information allocation is $\kappa^*(\theta)$, and the efficient incomplete-information allocation is

$$k^*(x, y) = \kappa_0^* + \kappa_1^*[(1 - \gamma^*)x + \gamma^*z],$$

where

$$\gamma^* = \delta + \frac{\alpha^* \delta (1 - \delta)}{1 - \alpha^* (1 - \delta)},$$

and

$$\alpha^* \equiv 1 - \frac{W_{KK}}{W_{\sigma\sigma}} = 1 - \frac{U_{kk} + 2U_{kK} + U_{KK}}{U_{kk} + U_{\sigma\sigma}}.$$

Interpretation: α^* is the socially optimal degree of coordination. It measures how much society wants agents to align actions, balancing the welfare cost of aggregate volatility against the welfare cost of dispersion.

4.6 Welfare-loss decomposition

For a given decentralized strategy, welfare can be written as

$$Eu = EW(\kappa^*, 0, \theta) + \frac{W_{KK}}{2} E(K - \kappa^*)^2 + \frac{W_{\sigma\sigma}}{2} E(k - K)^2.$$

Equivalently, welfare losses are

$$L^* = \frac{-W_{KK}}{2} \text{Var}(K - \kappa^*) + \frac{-W_{\sigma\sigma}}{2} \text{Var}(k - K).$$

Interpretation: The planner trades off volatility against dispersion. A higher α^* lowers dispersion but raises volatility. Whether that is good depends on which is more socially costly.

4.7 Equilibrium efficiency

The economy is efficient if and only if

$$\kappa(\theta) = \kappa^*(\theta) \quad \text{for all } \theta, \quad \alpha = \alpha^*.$$

Interpretation: There are two distinct efficiency margins. The complete-information level must be right, and the response to incomplete information must be right. Complete-information efficiency alone is not enough.

5 Theoretical design and identification logic

5.1 What is being identified?

The paper identifies the welfare consequences of information structures by reducing the problem to two comparisons:

- whether the complete-information equilibrium target κ equals the first-best target κ^* ;
- whether the equilibrium coordination motive α equals the socially optimal coordination motive α^* .

Interpretation: This makes the model a diagnostic tool. Once an applied environment is mapped into $\kappa, \kappa^*, \alpha, \alpha^*$, the welfare effects of public and private information can be understood.

5.2 Why the decentralized-information benchmark matters

A full-information planner would mechanically prefer more information. That comparison would not answer the paper's question. The paper holds information decentralized and asks whether agents respond to that information in a socially optimal way.

Interpretation: This is why the analysis can say that public information is bad in some economies without saying that ignorance is intrinsically valuable. Information can be harmful when it pushes decentralized equilibrium behavior in the wrong direction.

5.3 Why quadratic-Gaussian assumptions are used

- Quadratic payoffs make best responses and welfare losses linear/quadratic.
- Gaussian signals make conditional expectations linear.
- Together, they allow the entire equilibrium and planner allocation to be summarized by weights on x and z .

Interpretation: The assumptions are strong, but they make a very general welfare logic transparent: information affects accuracy and commonality; commonality affects volatility and dispersion.

5.4 Limitations of the theoretical design

- The model assumes a unique equilibrium. Multiple equilibria would add equilibrium-selection effects.
- Payoffs are concave. With local convexities, society might value volatility as a lottery.
- Information is exogenous. Endogenous information acquisition could interact with inefficiencies in information use.
- Communication among agents is ruled out by construction.

Interpretation: These restrictions are not incidental. They are what allow the paper to isolate efficient use of information rather than mixing it with equilibrium selection, experimentation, or communication design.

6 Section-by-section study guide

6.1 Sections 1–2: Motivation and model

The introduction explains why the positive implication of complementarity—more reliance on public information and more aggregate volatility—does not automatically imply a negative welfare judgment. Section 2 then sets up a linear-quadratic-Gaussian environment with a continuum of agents, private and public signals, and payoffs depending on individual actions, aggregate actions, dispersion, and fundamentals.

6.2 Section 3: Equilibrium use of information

This section derives Proposition 1. The key object is α , the private degree of coordination. The section shows that equilibrium public-signal weight γ rises with α . Complementarity therefore increases aggregate volatility and substitutability increases dispersion.

6.3 Section 4: Efficient use of information

This section introduces the decentralized team/planner problem. The central object is α^* , the socially optimal degree of coordination. The section shows that the efficient allocation has the same functional form as equilibrium but with α replaced by α^* and κ replaced by κ^* .

6.4 Section 5: Social value of information

This section classifies economies by the two efficiency gaps. It then studies the value of accuracy and commonality. The main lesson is that information is always valuable in efficient economies, but can reduce welfare when equilibrium uses information inefficiently or when the complete-information equilibrium is itself distorted.

6.5 Section 6: Applications

The paper maps the theory into competitive production, investment complementarities, beauty contests, new-Keynesian frictions, Cournot competition, and Bertrand competition. These applications show that similar-looking positive models can have opposite welfare implications.

7 Tables, figures, and theoretical result blocks

7.1 Proposition 1: Equilibrium Use of Information + Corollary 1: Volatility and Dispersion

Read:

- Proposition 1 characterizes equilibrium actions as a weighted average of private and public information.
- The equilibrium weight on public information is γ , which exceeds the Bayesian weight δ when actions are strategic complements.
- Corollary 1 translates this into observables: complementarity raises aggregate volatility and lowers dispersion.

Detailed interpretation: Complementarity increases the value of public information because public signals are common and therefore forecast other agents' actions. Substitutability does the reverse: private signals help agents differentiate. The positive equilibrium prediction is therefore clean even before welfare is considered.

Economic message: The result explains why coordination economies can overreact to public news and why substitute-action environments can display excessive cross-sectional dispersion.

Proposition 1: Equilibrium information weights

Equilibrium is

$$k(x, y) = \kappa_0 + \kappa_1[(1 - \gamma)x + \gamma z],$$

with

$$\gamma = \delta + \frac{\alpha\delta(1 - \delta)}{1 - \alpha(1 - \delta)}.$$

$\alpha > 0$ pushes agents toward public information; $\alpha < 0$ pushes them toward private information.

Corollary 1: Volatility and dispersion

Stronger complementarity decreases dispersion and increases nonfundamental aggregate volatility:

$$\frac{d \text{Var}(k - K)}{d\alpha} < 0 < \frac{d \text{Var}(K - \kappa)}{d\alpha}.$$

Public noise creates volatility; private noise creates dispersion.

7.2 Proposition 2: Efficient Use of Information + Corollary 2: Optimal Coordination

Read:

- Proposition 2 gives the efficient decentralized allocation.
- The planner's allocation has the same form as equilibrium, but with α^* replacing α .
- Corollary 2 explains the economic meaning of α^* : society wants more coordination when dispersion is costly and less coordination when volatility is costly.

Detailed interpretation: The planner cannot centralize information. The planner changes the rule mapping each agent's own signals to actions. Efficiency is therefore not about full information; it is about how to use partial information.

Economic message: The efficient amount of public-signal reliance is a welfare tradeoff between common mistakes and idiosyncratic mistakes.

Proposition 2: Efficient decentralized allocation

Efficient allocation is

$$k^*(x, y) = \kappa_0^* + \kappa_1^*[(1 - \gamma^*)x + \gamma^*z],$$

where γ^* is obtained from γ by replacing α with α^* .

Corollary 2: Optimal coordination

α^* decreases with social aversion to aggregate volatility and increases with social aversion to dispersion. A higher α^* means the planner wants agents to coordinate more strongly on common information.

7.3 Propositions 4–6: Efficient Economies and the Value of Information

Read:

- Proposition 4 gives the knife-edge condition for full efficiency: both $\kappa = \kappa^*$ and $\alpha = \alpha^*$.
- Proposition 5 shows that in efficient economies, accuracy is always valuable; commonality is valuable only when the efficient economy has complementarity.
- Proposition 6 shows that private and public information are both welfare-improving in efficient economies.

Detailed interpretation: The important lesson is that the Morris-Shin concern about public information cannot arise in an efficient economy. When equilibrium equals the decentralized planner's solution, the planner can always ignore extra information by adding noise, so information cannot reduce welfare.

Economic message: Public information is not harmful because it is public. It can be harmful only if the equilibrium response to common information is distorted.

Proposition 4: Efficiency condition

Economy is efficient iff

$$\kappa(\theta) = \kappa^*(\theta) \quad \forall \theta, \quad \alpha = \alpha^*.$$

Complete-information efficiency alone is not enough.

Propositions 5–6: Efficient economies

If $\kappa = \kappa^*$ and $\alpha = \alpha^*$, welfare increases with the precision of either private or public information. Complementarity raises the relative social value of public information.

7.4 Proposition 7 and Corollary 4: Inefficiency Only Under Incomplete Information

Read:

- These results cover economies where the complete-information target is correct but agents coordinate too much or too little under incomplete information.
- Accuracy remains welfare-improving because it reduces both volatility and dispersion.
- Commonality can be welfare-reducing if it pushes the economy toward the wrong mix of volatility and dispersion.

Detailed interpretation: If $\alpha > \alpha^*$, equilibrium is too coordinated. In that case, more common information may worsen welfare by further aligning actions. If $\alpha < \alpha^*$, equilibrium is insufficiently coordinated, and common information can be valuable.

Economic message: This is the cleanest formal version of the transparency tradeoff: public information can be bad when it increases an already-excessive tendency to coordinate.

Proposition 7: Incomplete-information inefficiency

When $\kappa = \kappa^*$ but $\alpha \neq \alpha^*$, welfare necessarily increases with accuracy σ^{-2} . The effect of commonality δ depends on whether coordination is too high or too low.

Corollary 4: Private vs. public precision

Public precision can reduce welfare only through the commonality channel. Private precision can reduce welfare only when less commonality is socially harmful.

7.5 Propositions 8–9 and Corollary 5: Inefficiency Even Under Complete Information**Read:**

- When $\kappa \neq \kappa^*$, more accurate information can push equilibrium actions closer to an inefficient complete-information target.
- The new object ϕ measures the relation between equilibrium activity and the first-best gap.
- If equilibrium activity moves in a direction correlated with the first-best distortion, more information can lower welfare.

Detailed interpretation: This is the paper’s strongest warning against simple information policy. Even accuracy, not merely commonality, can be bad when the complete-information equilibrium itself fluctuates inefficiently.

Economic message: Constructive ambiguity can be welfare-improving in distorted economies because less precise information may dampen movements toward inefficient complete-information allocations.

Propositions 8–9: Distorted complete information

If $\kappa \neq \kappa^*$, welfare losses include a first-order term involving the covariance between equilibrium deviations and the efficiency gap. Accuracy may be welfare-reducing.

Corollary 5: Sufficient conditions

When ϕ is sufficiently high, welfare increases with both private and public precision. When ϕ is sufficiently low, welfare decreases with both. The complete-information distortion governs the value of accuracy.

7.6 Application blocks: Competitive Production, Investment Complementarities, and Beauty Contests**Read:**

- Competitive production is efficient: substitutability creates dispersion, but the dispersion is socially appropriate.
- Investment complementarities create volatility, but the equilibrium may coordinate too little relative to the social optimum.
- Beauty contests differ because coordination is privately valuable but socially wasteful.

Detailed interpretation: These examples show why positive isomorphism is not enough for welfare analysis. Investment complementarities and beauty contests can both feature complementarity, but the social value of coordination differs.

Economic message: The model identifies when policy transparency is good or bad by asking whether coordination is socially valuable, not simply whether coordination is present.

Corollary 6: Competitive economy

Strategic substitutability raises dispersion, but equilibrium is efficient. Welfare increases with both private and public information.

Corollary 7 and 8: Investment vs. beauty contest

Investment complementarities: public information is welfare-improving because coordination is socially valuable. Beauty contests: public information can reduce welfare because coordination is socially undesirable.

7.7 Application blocks: Business Cycles, Cournot, and Bertrand

Read:

- Business-cycle transparency depends on the type of shock: information about efficient shocks helps; information about distortion shocks can hurt.
- In Cournot competition, information-sharing can reduce expected profits because firms coordinate quantities in a way that is not profit-maximizing.
- In Bertrand competition, information-sharing raises expected profits because pricing complementarities make common information beneficial.

Detailed interpretation: These examples demonstrate that welfare conclusions depend on the structure of distortions and strategic interaction. Public information is not uniformly pro- or anti-transparency.

Economic message: The same information policy can be desirable in one environment and harmful in another because it changes the correlation and accuracy of beliefs in different welfare-relevant ways.

Corollary 9: Business-cycle shocks

Information about the efficient source of the cycle raises welfare; information about the inefficient distortion source can lower welfare. Transparency should be shock-specific.

Corollaries 10–11: Cournot and Bertrand

Cournot: private information always raises expected total profits, but public information can lower them. Bertrand: both private and public information raise expected total profits.

8 Result map

- **Equilibrium map:** $\alpha \rightarrow \gamma \rightarrow$ reliance on public vs. private information.
- **Welfare map:** $\alpha^* \rightarrow \gamma^* \rightarrow$ socially optimal reliance on public vs. private information.
- **Inefficiency map:** compare α with α^* and compare κ with κ^* .
- **Information map:** decompose changes in signals into accuracy σ^{-2} and commonality δ .
- **Policy map:** public-information policy is desirable when it raises accuracy without inducing excessive commonality; it can be harmful when commonality worsens an existing coordination distortion.

9 Short summary

- The paper studies a large class of quadratic economies with dispersed private and public information.
- Strategic complementarity makes equilibrium actions more sensitive to public information; strategic substitutability makes them more sensitive to private information.
- Public noise creates aggregate volatility; private noise creates cross-sectional dispersion.
- The welfare analysis requires a decentralized-information efficiency benchmark.
- The efficient allocation has the same form as equilibrium but uses the socially optimal degree of coordination α^* .

- Equilibrium is efficient only if the complete-information target is efficient and $\alpha = \alpha^*$.
- In efficient economies, both private and public information raise welfare.
- If equilibrium uses information inefficiently, one type of information can lower welfare through its effect on commonality.
- If the complete-information equilibrium is distorted, even accuracy can reduce welfare.
- The applications explain why public information can be good in investment-complementarity and Bertrand environments but bad in beauty-contest and some Cournot or distortion-driven environments.

10 Conclusion

10.1 Core conclusion

Angeletos and Pavan show that the welfare value of information in coordination economies depends on the efficiency of the equilibrium use of information. Complementarity raises reliance on public information and aggregate volatility, but this is not necessarily socially excessive. Substitutability raises reliance on private information and cross-sectional dispersion, but that dispersion can be efficient.

10.2 Economic intuition

Public signals help agents coordinate because everyone observes the same thing. Private signals help agents tailor actions to fundamentals but make choices less aligned. The planner values both dimensions: alignment can be good when dispersion is costly, but it can be bad when common mistakes create costly aggregate volatility. The optimal mix depends on social preferences over volatility and dispersion, not simply on individual best responses.

10.3 Why the paper is important

The paper resolves a puzzle in the literature. Morris and Shin-style beauty contests suggested public information could be socially harmful, while other macro and coordination models found transparency beneficial. Angeletos and Pavan show that these results are not contradictions. The missing object is the social value of coordination, summarized by α^* , and the complete-information distortion summarized by $\kappa - \kappa^*$.

10.4 Policy implications

The framework cautions against universal prescriptions about transparency. Public disclosures by central banks, governments, or media can improve welfare when they mainly improve accuracy or help agents coordinate in a socially valuable way. But the same disclosures can lower welfare when they synchronize actions around socially undesirable responses or around distorted complete-information outcomes.

10.5 Limitations and next steps

The analysis relies on uniqueness, concavity, quadratic payoffs, Gaussian information, and exogenous information. Multiple equilibria, payoff convexities, communication, endogenous information acquisition, and market aggregation of private signals can introduce new welfare effects. The paper explicitly highlights these extensions as important directions for future work.

10.6 Final takeaway

The central lesson is: information is not socially valuable just because it is accurate, and public information is not harmful just because it is public. The welfare value of information depends on whether equilibrium uses decentralized information in the way society would like.